

Human vs. Physical Capital: *Explaining Income Differences*

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Introduction

This short portfolio study uses the 2019 cross-section of the Penn World Table (PWT) (v11.0) to examine how differences in physical capital per worker and human capital relate to cross-country differences in output per worker. Following standard growth-accounting intuition, I work in natural logs so coefficients can be read as elasticities and the relationships are approximately linear.

Data & Feature Engineering

The analysis uses variables from the **Penn World Table v11.0 (2019 cross-section)**. The dataset provides standardized measures of GDP, capital, human capital, and population across countries, allowing for international income comparisons.

Variable	Definition
rgdpna	Real GDP at constant 2021 national prices (in mil. 2021US\$)
rnna	Capital stock at constant 2021 national prices (in mil. 2021US\$)
emp	Number of persons engaged (in millions)
hc	Human capital index, based on years of schooling and returns to education

Figure 1. Variable Reference (Penn World Table v11.0)

From the dataset, I constructed the following variables:

- Output per worker (**Y**) = $\text{rgdpna} / \text{emp}$
- Capital per worker (**K**) = rnna / emp
- Human capital index (**Hc**) = **hc**

I then transformed the variables into natural logs for linear modeling:

- $\ln Y = \ln(Y)$
- $\ln K = \ln(K)$

- **$\ln H_c = \ln(H_c)$**

I then filtered the dataset to 2019 and dropped rows with missing values before running the regression models. The final dataset contains 143 countries after filtering.

Methods

I fit two ordinary least squares models: (1) a simple linear regression to establish the baseline relationship between capital and income, and (2) a multiple regression adding human capital as an additional covariate. Model performance is summarized using R^2 and RMSE.

1. Exploratory Visualization

Before modeling, I examined the relationship between capital per worker and output per worker. Figure 2 shows the scatter plot of $\ln Y$ vs $\ln K$ across countries in 2019.

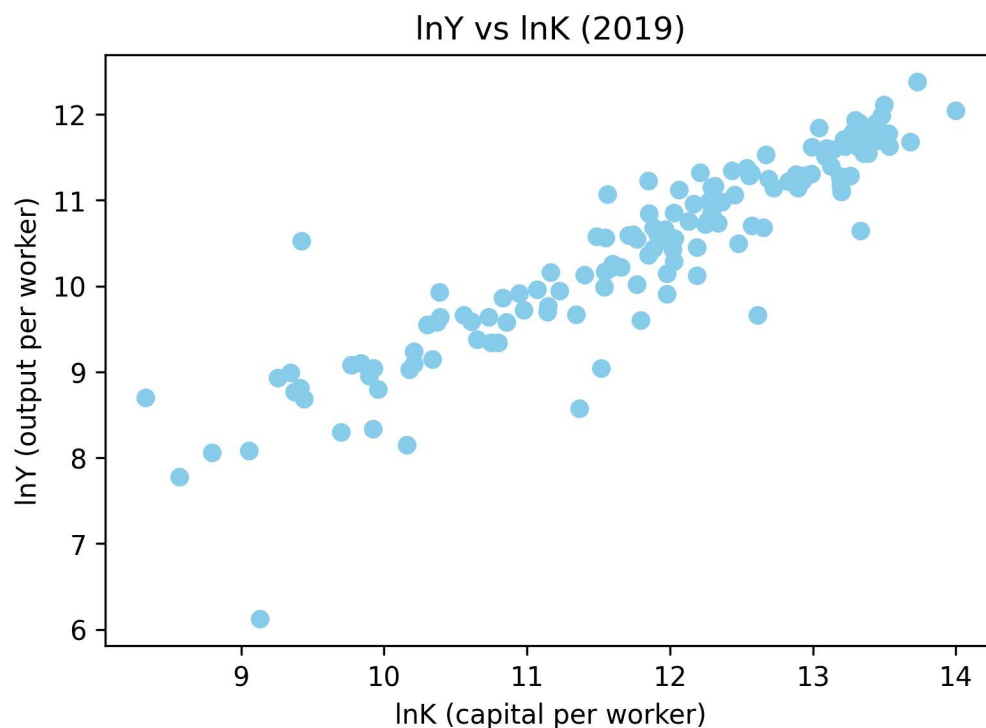


Figure 2. Scatter plot: $\ln Y$ vs $\ln K$ (2019).

2. Regression Analysis

I ran two regressions using scikit-learn's LinearRegression:

Simple Regression ($\ln Y$ on $\ln K$)

$$\ln Y = 0.771 \times \ln K + 1.325$$

$$R^2 = 0.840 \quad | \quad RMSE = 0.441$$

Multiple Regression ($\ln Y$ on $\ln K$ and $\ln H_c$)

$$\ln Y = 1.832 + 0.688 \times \ln K + 0.495 \times \ln Hc$$

$$R^2 = 0.846 \quad | \quad RMSE = 0.432$$

Figure 3 shows the fitted regression line over the data.

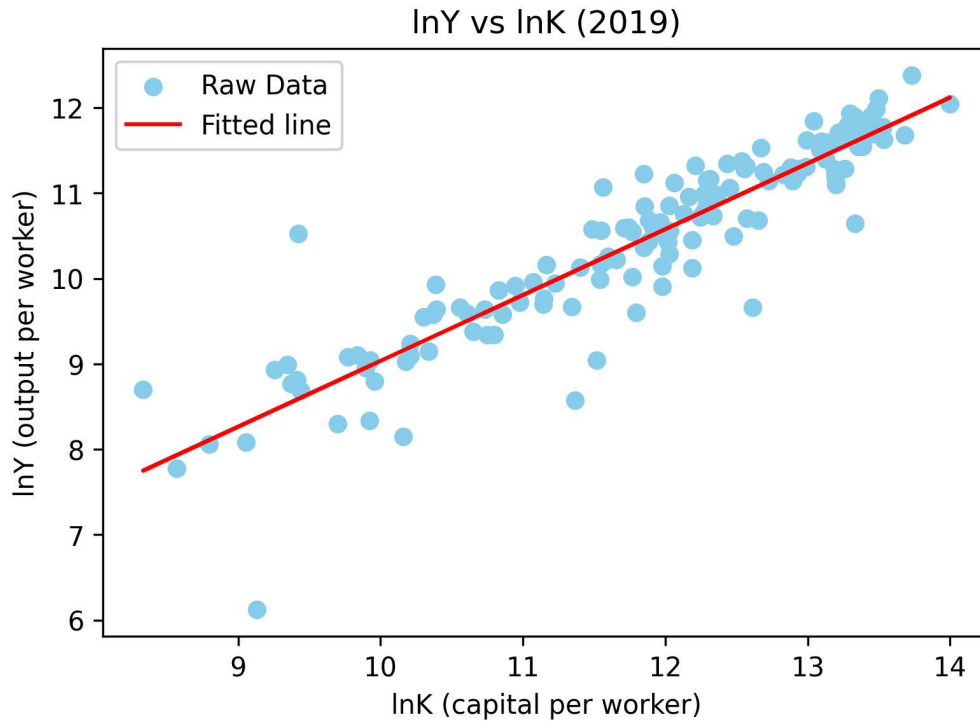


Figure 3. Linear fit: $\ln Y$ vs $\ln K$ (2019).

3. Residual Analysis

To identify countries where the model under- or over-predicts output per worker, I computed residuals (actual $\ln Y$ – predicted $\ln Y$) and ranked them by absolute size. The table in Figure 4 below lists the top 10 residuals from the multiple regression model.

Country	Actual $\ln Y$	Predicted $\ln Y$	Residual	Absolute Residual
Venezuela (Bolivarian Republic of)	6.121969	8.641759	-2.519790	2.519790
Syrian Arab Republic	10.524118	8.792045	1.732073	1.732073
Tajikistan	9.662277	11.082705	-1.420428	1.420428
Yemen	8.574039	9.957078	-1.383039	1.383039

Haiti	9.044084	10.038648	-0.994564	0.994564
Ukraine	10.644228	11.599555	-0.955327	0.955327
Central African Republic	8.149322	9.044187	-0.894865	0.894865
Egypt	11.070449	10.278196	0.792253	0.792253
Malawi	8.698014	7.919800	0.778214	0.778214
Gabon	11.225441	10.510753	0.714688	0.714688

Figure 4. Top 10 countries by absolute residual

Conclusion

This analysis shows that both physical capital and human capital play central roles in explaining cross-country differences in output per worker. Physical capital alone accounts for a substantial share of the variation ($R^2 \approx 0.84$), reflecting its fundamental importance in production. Incorporating human capital further improves the model's fit, capturing additional differences possibly linked to factors like education and skills.

Large residuals highlight cases where the model under- or over-predicts output, pointing to other factors—such as institutions, natural resource dependence, conflict, or measurement issues—that are not captured by the variables included here. Overall, this project strengthened my ability to transform economic data, apply regression methods, and interpret coefficients through the lens of growth theory.

Limitations & Next Steps: Future work could incorporate measures of institutional quality or resource dependence to better explain residual variation and refine model performance.

References

Feenstra, R. C., Inklaar, R., & Timmer, M. P. (2015). *Penn World Table version 11.0 (2019 cross-section)*. Groningen Growth and Development Centre, University of Groningen.
<https://www.rug.nl/ggdc/productivity/pwt>